256^3, z = 0: halo cross-correlation multipoles, mean over lines of sight x, y, z; mass bins 0: 2×10^{12}-10^{13}, 2: 5×10^{13}-2×10^{14}, 3: $>2\times10^{14}$ $M_\odot$/h; potential shift from ϕ_{r_s}

pair 0-0, $\ell=0$

pair 0-0, $\ell=1$

= 0 (by symmetry)

pair 0-0, $\ell=2$

pair 0-0, $\ell=4$

Gadget-4 (lines); pkdgrav3: circles
real
rsd
grav
rsd+grav

pair 0-2, $\ell=0$

pair 0-2, $\ell=1$

pair 0-2, $\ell=2$

pair 0-2, $\ell=4$

pair 0-3, $\ell=0$

pair 0-3, $\ell=1$

pair 0-3, $\ell=2$

pair 0-3, $\ell=4$

$s^2\xi_\ell$ [(Mpc/h)^2]

s [Mpc/h]